

Internal Cleaning Operations Platform

Developer Product Brief and Step-by-Step Build Plan

Objective: build one internal platform that replaces separate CRM, reservation, facility management, workforce coordination, SOP, communication, and financial reporting tools.

1. Executive Summary

This document defines the product vision, functional scope, workflows, data model, modules, implementation roadmap, and business value proposition for a custom internal platform for a cleaning services business.

The platform should centralize the complete operating model: lead capture, booking, job planning, employee assignment, SOP attribution, field execution, client communication, invoicing, payment tracking, and management reporting.

- Strategic goal: create a single source of truth for clients, sites, bookings, jobs, cleaners, invoices, payments, and operational history.
- Commercial goal: reduce administrative work, improve scheduling utilization, accelerate invoicing, and increase management visibility.
- Operational goal: standardize service delivery through checklists, SOPs, proof-of-service records, and mobile employee execution flows.
- Financial goal: produce real-time revenue, receivables, client profitability, and monthly performance dashboards from transactional data.

2. Value Proposition: Why Build Internally

Pain Point	Internal Platform Value	Business Impact
Fragmented systems	Unifies CRM, booking, task management, SOPs, communication, and finance in one product.	Lower tool cost, fewer manual handoffs, less operational leakage.
Manual invoicing and spreadsheet tracking	Connects completed tasks to invoice generation and payment status.	Faster billing cycle and improved cash visibility.
Difficult employee coordination	Mobile employee view shows assigned jobs, location, time window, SOP, checklist, and completion confirmation.	Fewer errors, better accountability, faster onboarding.
No centralized service history	Every client, site, task, issue, photo, invoice, and communication is stored in one timeline.	Better client service, stronger retention, easier dispute resolution.
Limited management reporting	Operational and financial dashboards provide revenue, utilization, unpaid invoices, recurring clients, and client concentration.	Better business decisions and scalable growth management.

3. Target Users and Roles

Role	Core Needs	Permissions
Owner / Admin	Full visibility across bookings, clients, employees, invoices, payments, reports, configuration, and user permissions.	Full access.
Operations Manager / Dispatcher	Create tasks, assign cleaners, monitor schedule, handle changes, view job progress, manage SOPs.	Operational access, limited financial controls.
Cleaner / Employee	View assigned tasks, SOPs, checklists, location, schedule, notes, and submit completion evidence.	Mobile-only execution access.
Finance / Accountant	View invoices, export reports, track payments, reconcile revenue, manage client billing details.	Financial access.

Client / Customer	Request service, approve quote, view booking, receive reminders, confirm completion, access invoices.	Client portal access.
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4. Core Product Modules

Module	Purpose	Key Features
CRM & Client Registry	Maintain all client, site, and contact information.	Client profiles, OIB/VAT ID, addresses, billing terms, service history, notes, files.
Reservation / Booking System	Capture and schedule one-time or recurring service requests.	Online booking, internal booking, recurring frequency, site selection, service type, quote estimate.
Task & Work Order Management	Convert bookings into operational tasks.	Task creation, status tracking, assignment, priority, due dates, dependencies, linked SOPs.
Employee Mobile View	Enable field teams to execute work cleanly.	Daily agenda, map/location, check-in/out, SOP checklist, photos, comments, proof-of-service.
SOP & Quality Control	Standardize service delivery.	SOP library, checklist templates, service-specific instructions, inspection results, issue logging.
Communication Funnel	Manage all client and employee communication.	Lead funnel, automated reminders, job updates, email/SMS/WhatsApp templates, internal comments.
Financial Reporting	Provide financial visibility and billing control.	Invoice generation, payment status, revenue dashboards, receivables, client revenue, exports.
Management Dashboard	Provide decision-ready operational insights.	Today schedule, jobs by status, cleaner utilization, revenue by month, overdue invoices, top clients.

5. End-to-End Operating Workflow

5.1 Booking Workflow

1. Lead or existing client submits a booking request through the website, client portal, phone entry by admin, or recurring contract setup.
2. System captures client, site address, service type, preferred date/time, frequency, estimated duration, special requirements, and attachments/photos if available.
3. System checks availability using employee capacity, existing jobs, travel buffers, and service duration rules.
4. Admin confirms booking, creates quote if needed, or converts directly into a scheduled job.
5. Client receives confirmation message with date, time window, service summary, and terms.
6. Booking automatically creates a work order and initial task record.

5.2 Task Creation Workflow

7. Work order is created from a booking, recurring contract, manual admin entry, or client request.
8. System assigns required service category: regular cleaning, deep cleaning, garage cleaning, carpet maintenance, window cleaning, drainage/emergency cleaning, office cleaning, etc.
9. Task stores client, site, service line, estimated time, required staff, required materials, price, due date, and operational notes.
10. System recommends SOP templates based on service type and site profile.
11. Task enters status: Draft, Scheduled, Assigned, In Progress, Completed, Reviewed, Invoiced, Paid.

5.3 Task Assignment Workflow

12. Dispatcher reviews task pipeline and daily calendar.
13. System shows available employees based on working hours, existing assignments, location, skills, and workload.
14. Dispatcher assigns one or multiple cleaners to the task.

15. Employee receives notification and sees the task in mobile agenda.
16. Employee can accept assignment or flag conflict if enabled.
17. Dispatcher monitors assignment status and can reassign if needed.

5.4 SOP Attribution Workflow

18. Each service category has default SOP templates and checklists.
19. When a task is created, the system automatically attaches the SOP relevant to service type, client, and site.
20. Admin can override or add custom instructions per client or site.
21. Employee must review SOP before starting and complete checklist before closing the job.
22. For high-risk or special jobs, mandatory photo evidence or supervisor approval can be required.
23. Completed SOP data is stored against the task and visible in client/service history.

5.5 Field Execution Workflow

24. Employee opens mobile app and views Today agenda.
25. Employee opens job card: client, address, time window, notes, SOP, checklist, materials, contact details.
26. Employee taps Check in when arriving on site; GPS/time stamp is stored.
27. Employee completes checklist items, uploads photos if required, logs issues, and adds comments.
28. Employee taps Complete; system captures end time, duration, checklist completion, issues, and evidence.
29. Task moves to Review or Completed depending on business rules.

5.6 Funnel Communication Workflow

30. Lead enters funnel through website form, phone entry, referral, existing client request, or imported contact.
31. Lead status moves through New Lead, Contacted, Quoted, Won, Scheduled, Active Client, Dormant, Lost.
32. System triggers communication templates at each stage: booking confirmation, quote follow-up, reminder, job completed, invoice sent, payment reminder.
33. All emails, SMS, WhatsApp messages, and internal notes are stored in the client timeline.
34. Admin can filter the funnel by source, status, expected value, follow-up date, and owner.
35. Management dashboard shows conversion rate, open quotes, won revenue, and lost reasons.

5.7 Financial Reporting Workflow

36. Completed job automatically becomes invoice-ready.
37. Admin reviews price, discount, VAT status, client billing details, and payment terms.
38. System generates invoice or exports invoice data to accounting software.
39. Payment status is tracked as Draft, Sent, Partially Paid, Paid, Overdue, Written Off.
40. Dashboard aggregates revenue by month, client, service type, employee/team, and payment status.
41. Reports can be exported to Excel/PDF for accountant, tax, or management review.

6. Functional Requirements

Area	Requirement	Priority
Client Management	Create, edit, search, and archive clients; store OIB, billing address, sites, contacts, terms, notes, files.	MVP
Booking	Create bookings manually and from web form; support one-time and recurring schedules.	MVP
Scheduling	Calendar and dispatch board with day/week/month view.	MVP
Task Assignment	Assign cleaners, set status, track capacity and job ownership.	MVP
Mobile Execution	Cleaner view with task list, check-in/out, checklist, notes, and photo upload.	MVP
SOP Library	Create SOP templates and attach to service types and tasks.	MVP
Communication	Email templates and notification logs; SMS/WhatsApp as later integrations.	Phase 2
Finance	Invoice-ready records, revenue dashboard, payment tracking, Excel export.	MVP
Reporting	Monthly revenue, top clients, unpaid invoices, job completion rate, employee utilization.	MVP

Client Portal	Client booking, job history, invoices, and service feedback.	Phase 2
Automation	Recurring jobs, auto reminders, payment reminders, late invoice alerts.	Phase 2
Advanced Analytics	Forecasting, profitability by client/service, route optimization.	Phase 3

7. Core Data Model

Entity	Key Fields	Relationships
Client	client_id, name, OIB/VAT ID, billing address, email, phone, payment terms, status	Has many Sites, Contacts, Bookings, Invoices.
Site	site_id, client_id, address, access notes, site type, preferred SOPs	Belongs to Client; has many Jobs.
Contact	contact_id, client_id, name, role, phone, email, communication preference	Belongs to Client.
Booking	booking_id, client_id, site_id, service type, date/time, recurrence, estimated price, status	Creates Jobs / Work Orders.
Job / Task	task_id, booking_id, client_id, site_id, assigned users, status, date, price, SOP, evidence	Belongs to Booking; has Assignments, Checklist Responses, Invoice Line.
Employee	employee_id, name, role, skills, availability, phone, status	Has Assignments and Timesheets.
SOP Template	sop_id, title, service type, checklist items, required photos, quality criteria	Attached to Service Types, Sites, and Tasks.
Invoice	invoice_id, invoice number, client, issue date, due date, amount, VAT, discount, status	Generated from completed Tasks.
Payment	payment_id, invoice_id, date, amount, method, reference	Belongs to Invoice.
Communication Log	message_id, channel, recipient, template, timestamp, status, related object	Linked to Client, Booking, Task, or Invoice.

8. Required Screens / Pages

Screen	Audience	Description
Dashboard	Owner / Admin	KPIs: revenue month-to-date, open jobs, completed jobs, unpaid invoices, overdue amount, top clients.
Client List	Admin / Finance	Search, filter, create, edit, view client history.
Client Profile	Admin / Finance	Client data, sites, contacts, jobs, invoices, messages, notes, documents.
Booking Form	Admin / Client	Create booking request with service, date, address, recurrence, price estimate.
Dispatch Calendar	Operations	Visual scheduling with drag-and-drop assignments and capacity overview.
Task Detail	Operations / Employee	Job summary, SOP, assigned staff, checklist, evidence, timeline, invoice linkage.
Employee Mobile Agenda	Employee	Today's tasks, check-in/out, SOP checklist, comments, upload photos.
SOP Library	Admin / Ops	Create and maintain SOP templates by service type.
Finance Dashboard	Owner / Finance	Revenue, open invoices, overdue invoices, client revenue, exports.
Funnel Board	Admin / Sales	Lead pipeline from inquiry to active client.

9. Recommended Technical Architecture

- Frontend web app: React / Next.js for admin, dispatcher, finance, and client portal.
- Mobile-first employee interface: responsive PWA initially; native app can follow later if needed.
- Backend API: Node.js/NestJS, Laravel, Django, or Ruby on Rails; choose based on developer capability and speed.
- Database: PostgreSQL as primary relational database.
- File storage: S3-compatible storage for job photos, attachments, invoices, and documents.
- Authentication: role-based access control with admin, dispatcher, employee, finance, and client roles.
- Notifications: email first; SMS and WhatsApp API integration as Phase 2.

- Reporting: SQL-based aggregation tables first; BI layer later if needed.
- Accounting export: Excel/CSV first; accounting system integration later.

10. MVP Roadmap

Phase	Scope	Outcome
Phase 0: Discovery	Map current workflows, invoice fields, client list, employee roles, service categories, and SOP templates.	Confirmed requirements and data structure.
Phase 1: Core Database + Admin App	Build clients, sites, bookings, tasks, employees, invoice-ready records, login/roles.	Operational backbone established.
Phase 2: Scheduling + Assignment	Calendar, dispatch board, employee assignment, status tracking, recurring bookings.	Business can manage daily operations in the platform.
Phase 3: Employee Mobile Execution	Mobile task view, SOP checklist, check-in/out, notes, photo upload, completion flow.	Field execution becomes standardized.
Phase 4: Financial Reporting	Invoice records, payment status, monthly revenue, client revenue, overdue invoices, Excel export.	Management can track cash and performance.
Phase 5: Communication Funnel	Lead pipeline, templates, automated reminders, client timeline.	Sales and client communication become systematic.
Phase 6: Client Portal + Automation	Online booking, client history, self-service invoice view, automated payment reminders.	Platform becomes client-facing and scalable.

11. Acceptance Criteria for MVP

- Admin can create a client, site, booking, task, assignment, SOP, and invoice-ready record without using Excel.
- Employee can log in from mobile, view assigned tasks, open SOP, check in, complete checklist, upload evidence, and mark task complete.
- Owner can view monthly revenue, unpaid invoices, top clients, completed jobs, and upcoming scheduled jobs.
- Dispatcher can manage schedule and assignments from one operational calendar.
- Finance user can export invoice data for accounting and track payment status.
- Every task has a clear audit trail: who created it, who was assigned, SOP used, completion time, evidence, and billing status.

12. Success Metrics

Metric	Why It Matters	Target Direction
Admin hours per week	Measures operational efficiency.	Decrease.
Booking-to-job conversion time	Measures scheduling speed.	Decrease.
Job completion accuracy	Measures SOP adherence and quality.	Increase.
Invoice cycle time	Measures speed from completed work to invoice.	Decrease.
Overdue invoice value	Measures cash discipline.	Decrease.
Revenue per client	Measures account value.	Increase.
Cleaner utilization	Measures workforce efficiency.	Increase without overloading staff.
Client repeat rate	Measures retention and service quality.	Increase.

13. Build vs Buy Conclusion

Buying separate CRM, reservation, field-service, facility-management, and financial tools may solve individual problems, but it creates data fragmentation and recurring subscription cost. Building internally can create a tailored operating system that matches the exact business model, invoice structure, service categories, employee workflows, and reporting needs.

The internal platform should start as a focused MVP, not a large enterprise system. The first version should prioritize operational control, employee execution, invoice-ready data, and management visibility. More advanced modules such as client portal, route optimization, WhatsApp automation, and profitability forecasting should be added only after the core workflow is stable.

14. Developer Handoff Checklist

- Confirm business roles and permissions.
- Confirm service categories and default SOPs.
- Import existing client and invoice dataset from Excel.
- Define invoice numbering rules and uniqueness logic.
- Design database schema and API endpoints.
- Build MVP screens: Dashboard, Clients, Bookings, Tasks, Calendar, Employees, SOPs, Finance.
- Build mobile employee task execution flow.
- Implement reporting exports and payment status tracking.
- Test with 10 real clients and 20 historical invoices before full rollout.
- Create admin training guide and SOP governance process.

15. Prioritized Product Backlog

Priority	Feature	User Story
P0	Client registry	As an admin, I need to store clients, OIB, addresses, contacts, and billing terms.
P0	Booking creation	As an admin, I need to create one-time and recurring bookings.
P0	Task assignment	As dispatcher, I need to assign cleaners and track status.
P0	Employee mobile view	As a cleaner, I need to see jobs, instructions, and checklists.
P0	SOP checklist	As manager, I need every job to follow a standard operating procedure.
P0	Invoice-ready jobs	As finance, I need completed jobs ready for invoicing.
P1	Financial dashboard	As owner, I need monthly revenue, unpaid invoices, and client revenue.
P1	Communication templates	As admin, I need booking confirmations and reminders.
P1	Client portal	As client, I want to request service and view invoices.
P2	Route optimization	As dispatcher, I want optimized daily routes.
P2	Advanced profitability	As owner, I want margin by client, service type, and employee team.

Prepared as a developer-ready product brief for building an internal cleaning operations platform.